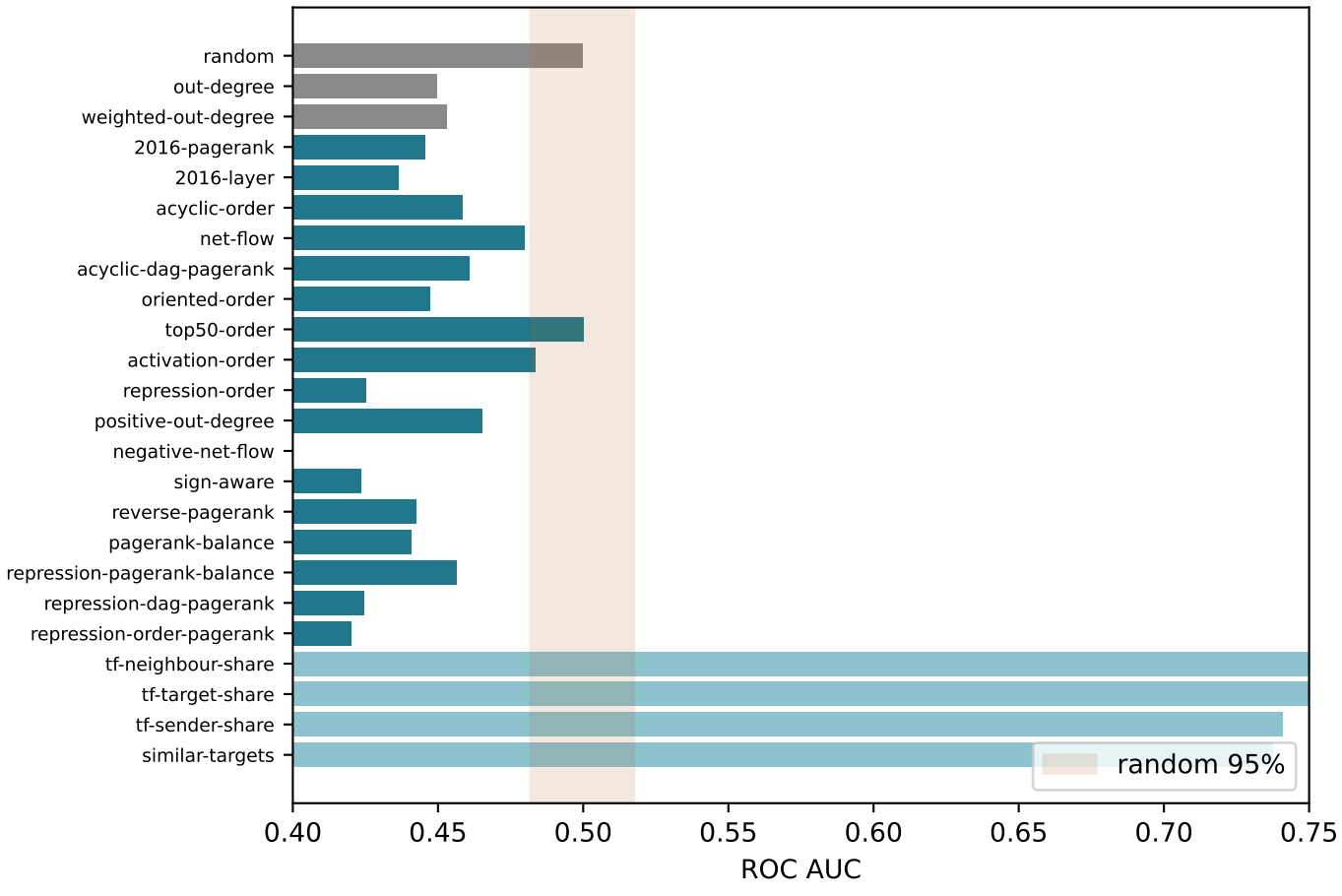
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,983 genes ranked, 1,153 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.6 (4–16)	0.500 (0.482–0.517)				
out-degree	10	0.450	0.497	1.000	-0.003 (-0.005, -0.002)	< 0.001
weighted-out-degree	12	0.453	0.249	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	11	0.446	0.360	1.000	-0.007 (-0.025, +0.010)	0.374
2016-layer	5	0.436	0.968	1.000	-0.017 (-0.033, -0.001)	0.042
acyclic-order	9	0.458	0.632	1.000	+0.005 (-0.002, +0.012)	0.164
net-flow	9	0.480	0.632	0.985	+0.027 (+0.014, +0.038)	< 0.001
acyclic-dag-pagerank	5	0.461	0.968	1.000	+0.008 (+0.002, +0.014)	0.008
oriented-order	9	0.447	0.632	1.000	-0.006 (-0.021, +0.009)	0.388
top50-order	5	0.500	0.968	0.486	+0.047 (+0.028, +0.066)	< 0.001
activation-order	11	0.484	0.360	0.963	+0.031 (+0.022, +0.039)	< 0.001
repression-order	5	0.425	0.968	1.000	-0.028 (-0.042, -0.014)	< 0.001
positive-out-degree †	13	0.465	0.159	1.000	+0.012 (+0.008, +0.016)	< 0.001
negative-net-flow †	1	0.396	1.000	1.000	-0.057 (-0.072, -0.043)	< 0.001
sign-aware †	3	0.423	0.999	1.000	-0.030 (-0.037, -0.022)	< 0.001
reverse-pagerank †	4	0.443	0.989	1.000	-0.010 (-0.014, -0.007)	< 0.001
pagerank-balance †	2	0.441	1.000	1.000	-0.012 (-0.026, +0.001)	0.084
repression-pagerank-balance †	7	0.457	0.870	1.000	+0.004 (-0.013, +0.018)	0.684
repression-dag-pagerank †	10	0.425	0.497	1.000	-0.028 (-0.043, -0.014)	< 0.001
repression-order-pagerank †	5	0.420	0.968	1.000	-0.033 (-0.046, -0.019)	< 0.001
tf-neighbour-share ‡	98	0.761	< 0.001	< 0.001	+0.308 (+0.285, +0.332)	< 0.001
tf-target-share ‡	93	0.752	< 0.001	< 0.001	+0.299 (+0.276, +0.324)	< 0.001
tf-sender-share ‡	98	0.741	< 0.001	< 0.001	+0.288 (+0.264, +0.312)	< 0.001
similar-targets ‡	96	0.738	< 0.001	< 0.001	+0.285 (+0.259, +0.310)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,874	0.039 (2.36×)	2.90×	0.64×	0.81 (0.81)
resolve_reciprocals → drop_sinks → prune_triangles	30,844	0.003 (0.75×)	2.31×	0.56×	0.40 (0.40)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	22,958	0.002 (0.64×)	1.80×	0.57×	0.33 (0.36)
absolute	31,874	0.039 (2.36×)			
absolute → minimum_feedback_order	31,668	0.028 (2.18×)			
orient_reciprocals ratio=2.0 → absolute	30,850	0.017 (1.82×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,782	0.017 (1.82×)			
top_outgoing k=50 → absolute	31,874	0.014 (3.61×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,532	0.009 (3.03×)			
keep_sign sign=1 → absolute	31,852	0.037 (2.90×)			
keep_sign sign=1 → absolute → minimum_feedback_order	31,666	0.025 (2.64×)			
keep_sign sign=-1 → absolute	30,186	0.003 (0.63×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	28,691	0.002 (0.59×)			
keep_sign sign=-1	30,186	0.003 (0.63×)		0.63×	0.17 (0.17)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,514,582 edges (619,013 negative), largest strong component 11,981.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,514,582 edges (619,013 negative), largest strong component 11,981.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 590,312 edges (533,134 negative), largest strong component 11,580.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 590,312 edges (533,134 negative), largest strong component 11,580; DAG keeps 326,744 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,514,582 edges (0 negative), largest strong component 11,981; DAG keeps 1,878,133 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,514,582 edges (0 negative), largest strong component 11,981.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,514,582 edges (0 negative), largest strong component 11,981; DAG keeps 1,878,133 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,367,522 edges (0 negative), largest strong component 10,590; DAG keeps 1,351,457 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 548,830 edges (0 negative), largest strong component 11,936; DAG keeps 394,252 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,895,569 edges (0 negative), largest strong component 11,969; DAG keeps 1,367,186 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 619,013 edges (0 negative), largest strong component 10,326; DAG keeps 521,901 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,514,582 edges (619,013 negative), largest strong component 11,981.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,514,582 edges (619,013 negative), largest strong component 11,981.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,514,582 edges (619,013 negative), largest strong component 11,981.

reverse-pagerank: Highest reverse PageRank (raw graph)